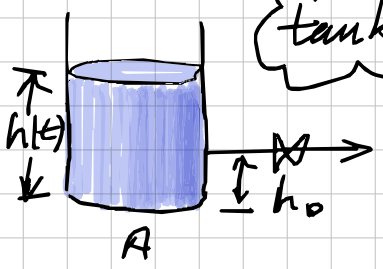
Differential energy balances



T_0

What is the relation?

tank drainage



energy balance:

accumulation = in − out + generation

$$U(t+\Delta t) - U(t) = -\dot{Q}(t)\Delta t \qquad t \to t+\Delta t$$

$$\Delta U = -\dot{Q}(t)\Delta t$$

$$\Delta U = MC_p\Delta T$$

$$MC_p\Delta T = -\dot{Q}(t)\Delta t$$

$$\Delta T = -\frac{\dot{Q}(t)}{MC_p}\Delta t$$

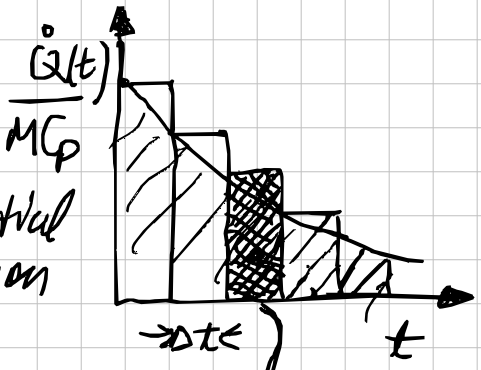
$$\dot{Q}(t) = hS(T - T_0)$$

$$\Delta T = -\frac{hS(T(t) - T_0)}{MC_p}\Delta t$$

$$\frac{\Delta T}{T - T_0} = -\frac{hS}{MC_p}\Delta t$$

analytical solution

$$\int_{T(0)}^{T(t)} \frac{dT}{T - T_0} = -\int_0^t \frac{hS}{MC_p}\,dt$$



the area of each rectangle is equal to $\Delta T(t) = T(t+\Delta t) - T(t)$

$$T(t+\Delta t) - T(t) = -\frac{\dot{Q}(t)}{MC_p}\Delta t$$

$$\int_{T(0)}^{T(t)} \frac{dT}{T - T_0} = -\int_0^t \frac{hS}{MC_p}\, dt$$

analytical solution ←

$$\ln \frac{T(t) - T_0}{T(0) - T_0} = -\frac{hS}{MC_p} t = -t/\tau$$

where $\tau = \frac{MC_p}{hS}$

$T(0)$ = initial temperature of object.

$$T(t) = T_0 + (T(0) - T_0)\, e^{-t/\tau}$$

Euler's method: numerical solution

$$T(t+\Delta t) - T(t) \simeq -\frac{\dot{Q}(t)}{MC_p} \Delta t \quad \leftarrow \text{time step}$$

$$T(t+\Delta t) = T(t) - \frac{\dot{Q}(t)}{MC_p} \Delta t$$

$$= T(t) - \frac{hS}{MC_p} [T(t) - T_0] \Delta t$$

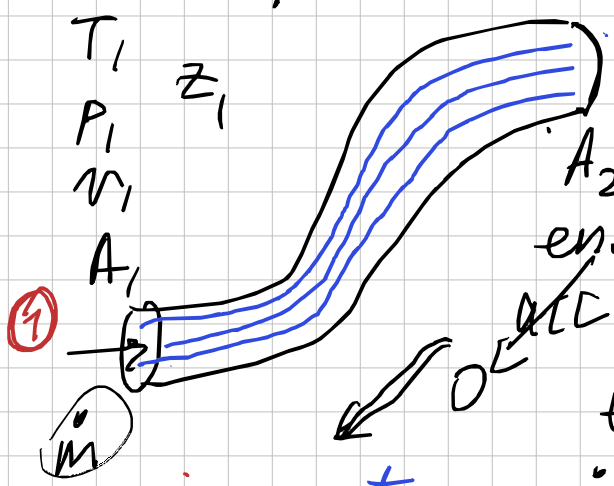
$$= T(t) - [T(t) - T_0] \frac{\Delta t}{\tau}$$

How do things change if the heat capacity depends on temperature, such as

$C_p(T) = 4.2 + 0.001\, T$?

$$f(x + \Delta x) \simeq f(x) + f'(x)\, \Delta x$$

Bernoulli's equation



energy balance

$\Delta E = in - out + gen^{0}$

① ②

$t \to t + \Delta t$

$\dot{m} = \rho A_1 v_1 = \rho A_2 v_2$

in = out

$(\rho A_1 v_1)\Delta t \frac{v_1^2}{2}$

$+ P_1 A_1 v_1 \Delta t$

$+ (\rho A_1 v_1)\Delta t\, g z_1$

$+ [\rho A_1 v_1)\Delta t\, u_1 = (\rho A_2 v_2)\Delta t \frac{v_2^2}{2}$

$+ P_2 A_2 v_2 \Delta t$

$+ \rho A_2 v_2 \Delta t\, g z_1$

$+ [\rho A_2 v_2)\Delta t\, u_2$

$\Delta W = F \Delta x$

$= \frac{F}{A}(A \Delta x)$

$= P \Delta V$

$\dot{m}\Delta t \frac{v_1^2}{2} + P_1 \frac{\dot{m}}{\rho}\Delta t + \dot{m}\Delta t g z_1 + \dot{m}\Delta t u_1$

$= \dot{m}\Delta t \frac{v_2^2}{2} + P_2 \frac{\dot{m}}{\rho}\Delta t + \dot{m}\Delta t g z_2 + \dot{m}\Delta t u_2$

$\frac{v_1^2}{2} + \frac{P_1}{\rho} + g z_1 + u_1 = \frac{v_2^2}{2} + \frac{P_2}{\rho} + g z_2 + u_2$

What is enthalpy?

$H = U + pV$ ← definition

$$\frac{v^2}{2} + \underline{u + \frac{p}{\rho}} + gz = \text{const}$$

$u + pV$

$V = 1/\rho$ specific volume.

$$\frac{v^2}{2} + H + gz = \text{const}$$ ← generalized Bernoulli's equation.

↑ enthalpy